

tokens $B \times T$

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graph LR; A[tokens B x T] --> B["Q K V: B x H x T x D"]; B --> C[masked attention]; C --> D["MLP + residual"]; D --> E[next-token / head];
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The diagram illustrates the architecture of a transformer block. It begins with an input of 'tokens $B \times T$ ' enclosed in a dashed box. This input is processed through a series of four components: 1) A blue box containing 'Q K V: $B \times H \times T \times D$ ', which represents the projection of input tokens into query, key, and value matrices. 2) A blue box labeled 'masked attention', representing the self-attention mechanism. 3) A blue box labeled 'MLP + residual', representing the multi-layer perceptron with a residual connection. 4) A green box labeled 'next-token / head', representing the output of the block. Blue arrows connect the first three components, while a green arrow connects the final component to the output.

Q K V: $B \times H \times T \times D$

masked attention

MLP + residual

next-token / head